6(i)

i) Walrasian demand f^n:-

Walrasian demand is a set of consumption bundle which maximises consumers utility given a budget constraint for given level of prices and ~~uti~~ wealth.

$$\max \; u(x_1, x_2)$$
$$\text{s.t.} \; px \leq \omega$$

~~Now~~, Derivation of walrasian demand f^n:-

Given, $u(x_1, x_2) = K x_1^{\alpha} x_2^{1-\alpha}$

s.t. $p_1 x_1 + p_2 x_2 = \omega_0$

Now, $L = K x_1^{\alpha} x_2^{1-\alpha} + \lambda \left[\omega_0 - p_1 x_1 - p_2 x_2 \right]$

FOC ~~requires~~,

$$\cancel{\frac{\delta L}{\delta P_1} = 0}$$

Now, FOC requires,

$$\frac{\delta L}{\delta x_1} = 0 \Rightarrow \alpha k x_1^{\alpha-1} x_2^{1-\alpha} - \lambda P_1 = 0$$

$$\Rightarrow \alpha k x_1^{\alpha-1} x_2^{1-\alpha} = \lambda P_1 \quad \ldots (i)$$

$$\frac{\delta L}{\delta x_2} = 0 \Rightarrow (1-\alpha) k x_1^{\alpha} x_2^{1-\alpha-1} - \lambda P_2 = 0$$

$$\Rightarrow (1-\alpha) k x_1^{\alpha} x_2^{-\alpha} = \lambda P_2 \quad \ldots (ii)$$

$$\frac{\delta L}{\delta \lambda} = P_1 x_1 + P_2 x_2 = \omega \quad \ldots (iii)$$

Now, (i) ÷ (ii) we get,

$$\frac{P_1}{P_2} = \frac{\alpha k x_1^{\alpha-1} x_2^{1-\alpha}}{(1-\alpha) k x_1^{\alpha} x_2^{-\alpha}}$$

$$= \frac{\alpha x_1^{\alpha-1} x_2^{1-\alpha}}{(1-\alpha) x_1^{\alpha} x_2^{-\alpha}}$$

$$= \frac{\alpha}{(1-\alpha)} x_1^{\alpha-1-\alpha} \cdot x_2^{1-\alpha+\alpha}$$

$$= \frac{\alpha x_2}{(1-\alpha) x_1}$$

$$\therefore \frac{p_1}{p_2} = \frac{\alpha x_2}{(1-\alpha)x_1}$$

$$\Rightarrow \alpha x_2 = \frac{(1-\alpha)p_1 x_1}{p_2}$$

$$\Rightarrow x_2 = \frac{(1-\alpha)p_1 x_1}{\alpha p_2} \quad \ldots (iv)$$

Now, putting x_2 from (iv) in (iii), we g

$$p_1 x_1 + p_2\left[\frac{(1-\alpha)p_1 x_1}{\alpha p_2}\right] = \omega$$

$$\Rightarrow p_1 x_1 + \frac{(1-\alpha)}{\alpha} p_1 x_1 = \omega,$$

$$\cancel{\Rightarrow p_1 x_1 \left[1 + 1 - \alpha\right] = \omega_1}$$

$$\Rightarrow p_1 x_1 \left[1 + \frac{1-\alpha}{\alpha}\right] = \omega$$

$$\Rightarrow p_1 x_1 \left[\frac{\alpha + 1 - \alpha}{\alpha}\right] = \omega$$

$$\Rightarrow p_1 x_1 \left(\frac{1}{\alpha}\right] = \omega.$$

$$\Rightarrow x_1 = \frac{\alpha \omega}{p_1} \quad \ldots (v)$$

Putting x_1 in (iv), we get,

$$x_2 = \frac{(1-\alpha) P_1 \left[\frac{\alpha \omega}{P_1}\right]}{\alpha P_2}$$

$$= \frac{(1-\alpha)\omega}{P_2}$$

∴ Walrasian Demand f^n's are:–

$$(x_1, x_2) = \left[\frac{\alpha \omega}{P_1}, \frac{(1-\alpha)\omega}{P_2}\right]$$

Properties:-

① Homogenous of degree 0:-

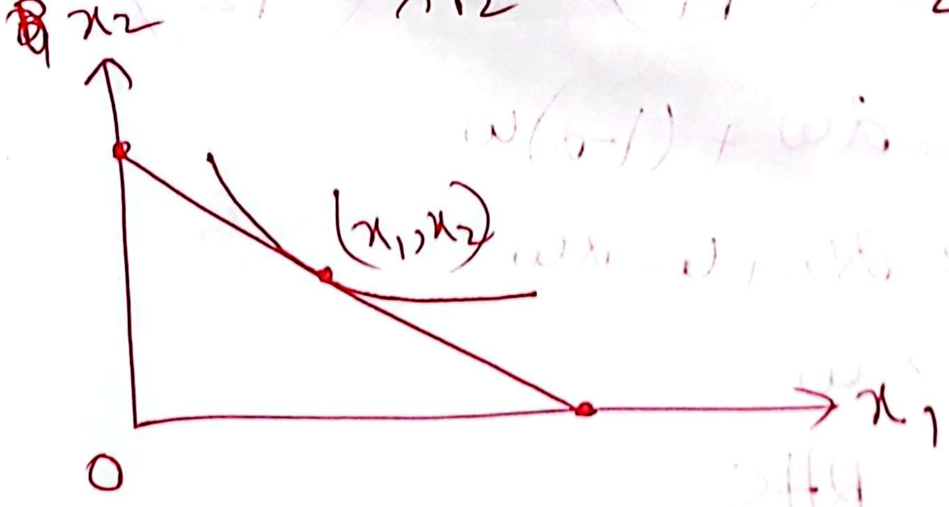
Now, it means that if the wealth and prices are increased by same proportion then it will not alter the actual consumption bundle.

Proof:-

Increasing ω by λ and P_1, P_2 by λ, we g

$$x_1 = \frac{\alpha \lambda \omega}{\lambda . P_1} = \frac{\alpha \omega}{P_1}$$

$$x_2 = \frac{(1-\alpha)\lambda \omega}{\lambda P_2} = \frac{(1-\alpha)\omega}{P_2}$$



$\boxed{\text{(2) Walras law:-}}$

Walras law indicates that

$$px = \omega.$$

i.e. if the new consumption bundles are put in the budget constraint then it will be equal to the initial level of wealth.

$$L.H.S = P_1 x_1 + P_2 x_2$$

$$= P_1\left(\frac{\alpha\omega}{P_1}\right) + P_2\left(\frac{(1-\alpha)\cdot\omega}{P_2}\right)$$

$$= \alpha\omega + (1-\alpha)\omega$$

$$= \alpha\omega + \omega - \alpha\omega$$

$$= \omega$$

$$= RHS$$

[Proved]

③ Convexity:-

~~The convex combination of~~

If the preferences are convex, then the walrasian demand must also be convex

If it is strictly convex, then the walrasian demand will also be strictly convex.

The combinations of any ~~two~~ bundles in the set will also be in the set.

6(ii)

Indirect utility f^n:-

If we put the walrasian demand functions in the utility function then the new utility function obtained will be indirect utility f^n.

From (i), we get;

$$x_1 = \frac{\alpha w}{p_1}, \quad x_2 = \frac{(1-\alpha)w}{p_2}$$

Now, ~~we~~ given, $U(x_1, x_2) = K x_1^{\alpha} x_2^{1-\alpha}$

putting x_1 and x_2 in $U(x_1, x_2)$, we get,

$$v = k\left(\frac{\alpha w}{p_1}\right)^{\alpha}\left[\frac{(1-\alpha)w}{p_2}\right]^{1-\alpha}$$

$$= k\,\frac{\alpha^{\alpha}w^{\alpha}}{p_1^{\alpha}}\cdot\frac{(1-\alpha)^{1-\alpha}w^{1-\alpha}}{p_2^{1-\alpha}}$$

$$= k\,\frac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}w^{\alpha+1-\alpha}}{p_1^{\alpha}p_2^{1-\alpha}}$$

$$= k\,\frac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}w}{p_1^{\alpha}p_2^{1-\alpha}}$$

$$\therefore\ v(p,w) = kw\,\frac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{p_1^{\alpha}p_2^{1-\alpha}}$$

~~Pro~~ $\boxed{\text{Properties:-}}$

1. Homogenous of degree 0 in prices
2. Strictly increasing in w, but non increasing in P
3. Quasi-convexity → Book

① Homogeneous of degree 0 in ~~prices~~

If we ~~ef~~ increase the prices by λ, then there will be no change in the indirect utility f^n.

Now, $v(p,\omega) = K\omega \dfrac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{p_1^{\alpha} p_2^{1-\alpha}}$

Now, $v(\lambda p, \lambda\omega) = K(\omega\lambda) \dfrac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{(\lambda p_1)^{\alpha}(\lambda p_2)^{1-\alpha}}$

$= K\omega \lambda \dfrac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{\lambda^{\alpha+1-\alpha}}$

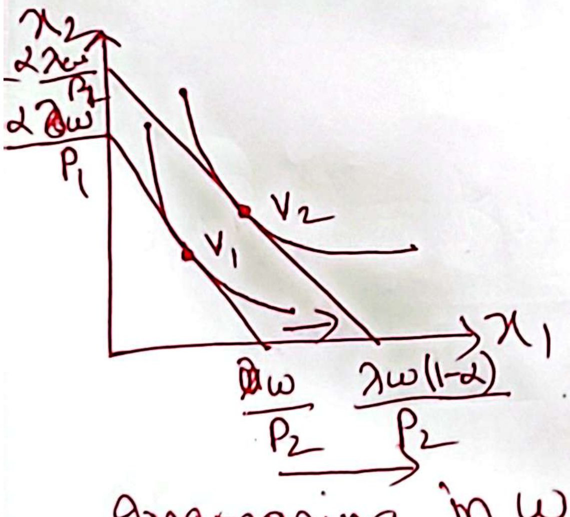
$= K\omega \lambda \dfrac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{\lambda p_1^{\alpha} p_2^{1-\alpha}}$

$= K\omega \dfrac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{p_1^{\alpha} p_2^{1-\alpha}}$

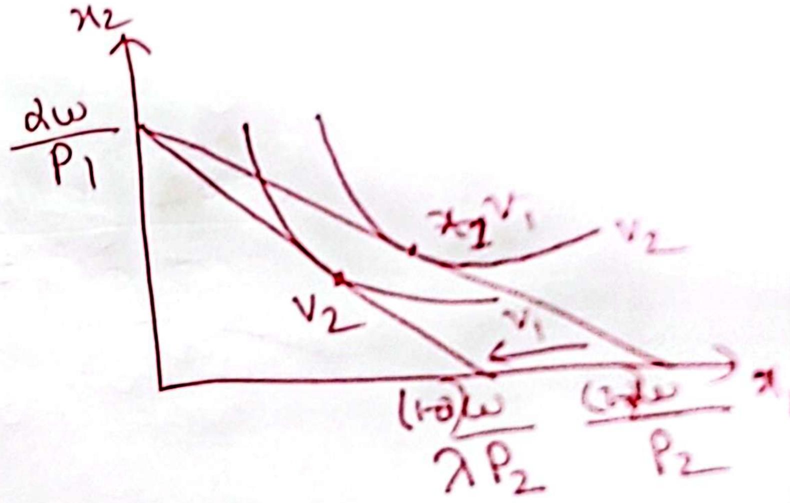
$= v(p,\omega)$.

② Strictly increasing in ω and non increasing in p:-

If the wealth level is increased then the budget constraint will shift rightward i.e. the ~~utility~~ indirect utility function will increase.



Increasing in ω.



Non increasing in p.

$-(1-\alpha)-1 \Rightarrow \alpha-1-1 \Rightarrow \alpha-2$

Roy's Identity:

$$\underset{\cancel{(x_1, x_2)}}{x_i(p,\omega)} = -\frac{\partial v / \partial p_i}{\partial v / \partial \omega}$$

i.e. $x_1 = \dfrac{-\partial v / \partial p_1}{\partial v / \partial \omega}$ and $x_2 = \dfrac{-\partial v / \partial p_2}{\partial v / \partial \omega}$

Now, $v(p,\omega) = k\omega \dfrac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha \, p_2^{1-\alpha}}$

Now, $\dfrac{\partial v}{\partial p_1} = k\omega \dfrac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_2^{1-\alpha}} \cdot (-\alpha) \, p_1^{-\alpha-1}$

$$= -\alpha k\omega \frac{\alpha^\alpha (1-\alpha)^{1-\alpha} \, p_1^{-\alpha-1}}{p_2^{1-\alpha}}$$

$\dfrac{\partial v}{\partial p_2} = k\omega \dfrac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha} \cdot (-(1-\alpha)) \, p_2^{-1+\alpha-1}$

$$= -(1-\alpha)(k\omega) \frac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha} \, p_2^{\alpha-2}$$

$\dfrac{\partial v}{\partial \omega} = k \dfrac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha \, p_2^{1-\alpha}}$

Now, $x_1 = \frac{-\partial P \cdot \partial v/\partial p_1}{\partial v/\partial \omega}$

$$= \frac{-\left(-\alpha k \omega \frac{\alpha^\alpha (1-\alpha)^{1-\alpha} p_1^{-\alpha-1}}{p_2^{1-\alpha}}\right)}{k \frac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha p_2^{1-\alpha}}}$$

$\therefore x_1 = \frac{\alpha \omega}{p_1}$

And, $x_2 = \frac{-\partial v/\partial p_2}{\partial v/\partial \omega}$

$$= \frac{-\left(-(1-\alpha) k \omega \frac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha} p_2^{\alpha-2}\right)}{\frac{k \alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha p_2^{1-\alpha}}}$$

$\therefore x_2 = \frac{(1-\alpha)\omega}{p_2}$

Thus Roy's Identity is verified.

6(iii)

Expenditure minimization problem:-

This means minimizing cost for a given utility level by a bundle of consumption set X. This type of demand f^n is known as compensated demand function or Hicksian demand function

i.e. $\min \; px = \omega$.

s.t. $u(x_1, x_2)$

Solving expenditure minization problem:

min

$p_1x_1 + p_2x_2 = \omega$

s.t. $u(x_1,x_2) = kx_1^{\alpha}x_2^{1-\alpha}$

Now, $L = (p_1x_1 + p_2x_2 - \omega) + \lambda\left[kx_1^{\alpha}x_2^{1-\alpha}\right]$

FOC requires -

$$\frac{\delta L}{\delta x_1} = p_1 + k\lambda x_2^{1-\alpha}(\alpha)x_1^{\alpha-1} = 0$$

$$\Rightarrow p_1 = -\alpha\lambda k x_1^{\alpha-1}x_2^{1-\alpha} \quad \ldots(i)$$

$$\frac{\delta L}{\delta x_2} = p_2 + \lambda k(1-\alpha)x_2^{1-\alpha-1}x_1^{\alpha} = 0$$

$$= p_2 = -\lambda k(1-\alpha)x_2^{-\alpha}x_1^{\alpha} \quad \ldots(ii)$$

$$\frac{\delta L}{\delta \lambda} = kx_1^{\alpha}x_2^{1-\alpha} = \cancel{0}u \quad \ldots(iii)$$

(i) ÷ (ii), we get,

$$\frac{P_1}{P_2} = \frac{-\alpha\lambda K x_1^{\alpha-1} x_2^{1-\alpha}}{-\lambda K(1-\alpha)\, x_2^{-\alpha} x_1^{\alpha}}$$

$$= \frac{\alpha}{1-\alpha}\, x_1^{\alpha-1+\alpha}\, x_2^{1-\alpha+\alpha}$$

$$\frac{P_1}{P_2} = \frac{\alpha x_2}{(1-\alpha)x_1}$$

$$\Rightarrow x_2 = \frac{P_1 x_1 (1-\alpha)}{P_2 \alpha} \quad \ldots (iv)$$

Putting (iv) in (iii), we get,

$$K x_1^{\alpha}\left[\frac{P_1 x_1 (1-\alpha)}{P_2 \alpha}\right]^{1-\alpha} = \phi\, U$$

$$\Rightarrow K x_1^{\alpha}\, x_1^{1-\alpha}\left[\frac{(1-\alpha)P_1}{\alpha P_2}\right]^{1-\alpha} = \phi\, U.$$

$$\Rightarrow K x_1^{\alpha+1-\alpha}\left[\frac{(1-\alpha)P_1}{\alpha P_2}\right]^{1-\alpha} = \phi\, U$$

$$\Rightarrow x_1 = \frac{U}{K}\left(\frac{\alpha P_2}{(1-\alpha)P_1}\right)^{1-\alpha}$$

$$x_2 = \frac{P_1 x_1 (1-\alpha)}{\alpha P_2}$$

$$= \frac{P_1(1-\alpha)\left[\frac{U}{k}\left(\frac{\alpha P_2}{(1-\alpha)P_1}\right)^{1-\alpha}\right]}{\alpha P_2}$$

$$= \frac{U}{k}(1-\alpha)^{1-1+\alpha} \cdot P_1^{1-1+\alpha} \cdot \alpha^{1-\alpha-1} \cdot P_2^{1-\alpha-1}$$

$$= \frac{U}{k}(1-\alpha)^{\alpha} \cdot P_1^{\alpha} \cdot \alpha^{-\alpha} \cdot P_2^{-\alpha}$$

$$= \frac{U}{k}\left[\frac{(1-\alpha)P_1}{\alpha P_2}\right]^{\alpha}$$

$$\therefore x_1 = \frac{U}{k}\left(\frac{\alpha P_2}{(1-\alpha)P_1}\right)^{1-\alpha}$$

$$x_2 = \frac{U}{k}\left(\frac{(1-\alpha)P_1}{\alpha P_2}\right)^{\alpha}$$

Characteristics:-

① Homogeneity of degree 0 in prices:-

If Prices are increased by λ, then there will be no change in the compensated demand function.

$$\cancel{\text{i.e. } x_1(\lambda P, U) = \frac{U}{k}\left[\frac{(1-\alpha)\lambda P_1}{\alpha \lambda P_2}\right]}$$

i.e. $$x_1(\lambda P, U) = \frac{U}{k}\left[\frac{\alpha \lambda P_2}{(1-\alpha)\lambda P_1}\right]^{1-\alpha}$$

$$= \frac{U}{k}\left[\frac{\alpha P_2}{(1-\alpha) P_1}\right]^{1-\alpha}$$

$$= x_1(P, U)$$

and, $$x_2(\lambda P, U) = \frac{U}{k}\left[\frac{(1-\alpha)\lambda P_1}{\alpha \lambda P_2}\right]^{\alpha}$$

$$= \frac{U}{k}\left[\frac{(1-\alpha) P_1}{\alpha P_2}\right]^{\alpha}$$

$$= x_2(P, U)$$

② No excess utility :-

If the ~~com~~ optimal bundle is put in the existing ~~utility~~ utility function, then there will be no excess utility.

Putting x_1 and x_2 in $U(x_1, x_2)$ we get,

$$U(x_1, x_2) = K x_1^{\alpha} x_2^{1-\alpha}$$

$$= K\left[\frac{U}{K}\left[\frac{\alpha P_2}{(1-\alpha)P_1}\right]^{1-\alpha}\right]^{\alpha}\left[\frac{U}{K}\left(\frac{(1-\alpha)P_1}{\alpha P_2}\right)^{\alpha}\right]^{1-\alpha}$$

$$= K\left(\frac{U}{K}\right)^{\alpha+1-\alpha}\left[\frac{\alpha P_2}{(1-\alpha)P_1}\right]^{\alpha-\alpha^2}\left[\frac{(1-\alpha)P_1}{\alpha P_2}\right]^{\alpha-\alpha^2}$$

= U

= no ex com utility.

③ ~~Uniquenesses~~ Uniqueness:-

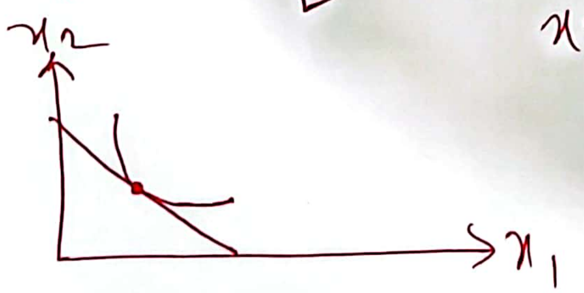
If ~~the optimal bundle~~ preference relation is strictly convex, then h(p,w) contains a single element.

④ Convexity:- If preference relation is convex then h(p,w) is also convex set.

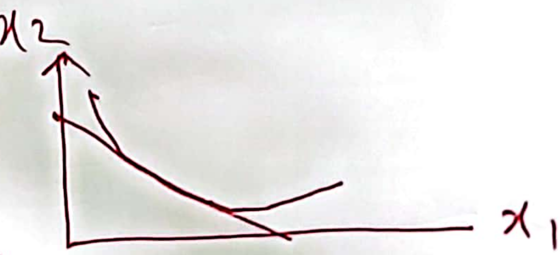
⑤ Law of compensated demand:-

If $p' > p$, then,

$[p'-p]\left[h(p',U) - h(p,U)\right] \leq 0$ holds.



Uniqueness.



Convexity